MATH 122 — Calculus II · Exam 2 Makeup: Techniques of Integration

Name: _____ **Date:** _____

Instructions:

- Show all work: Full credit requires complete, correct steps and proper math notation.
- Points: Each question is worth 10 points.
- Formula Sheet: Permitted (formulas only; no worked examples).
- Calculators: Basic, non-graphing calculators allowed.

Part I: Integration by Parts

1. (10 points) Evaluate

$$\int x^{15} \ln x \, dx$$

2. (10 points) Evaluate

$$\int 4x^3 e^{-x} \, dx$$

3. (10 points) Evaluate

$$\int \tan^{-1}(x)\, dx$$

4. (10 points) Evaluate the definite integral

$$\int_0^{\pi/2} x \cos x\, dx$$

Part II: Trigonometric Integrals

5. (10 points) Evaluate

$$\int \cos^7 x \, dx$$

6. (10 points) Evaluate

$$\int \sin^2 x \cos^3 x \, dx$$

7. (10 points) Evaluate

$$\int \sin 8x \cos 3x \, dx$$

8. (10 points) Evaluate

$$\int 4 \sin 2x \cos 4x \cos 6x \, dx$$

Part III: Partial Fractions

9. (10 points) Evaluate

$$\int \frac{6}{(x-1)(x+2)}\,dx$$

10. (10 points) Evaluate

$$\int \frac{9}{x^2(x-3)}\,dx$$